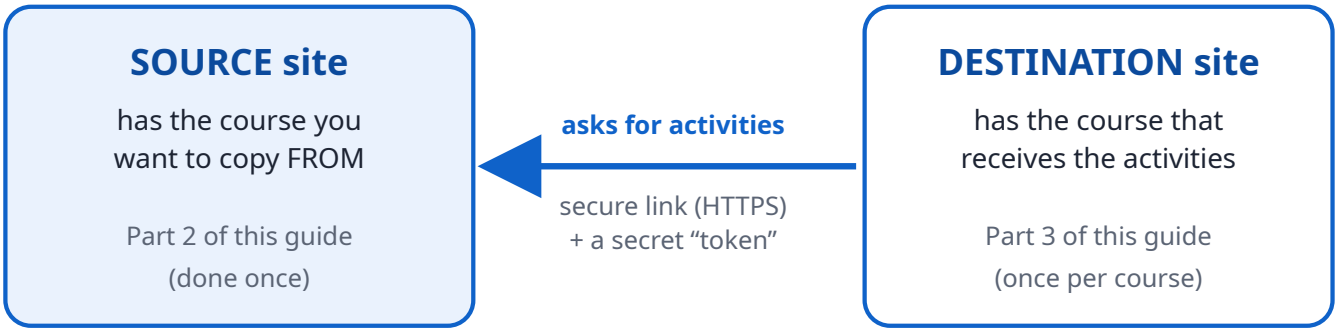
COURSE SYNC FOR MOODLE

Set-up Walk-Through

A step-by-step guide for administrators who work in the Moodle web screens. Every step is done with a mouse and a keyboard: no command line, no server access.



Part 1	Install the plugin on both sites
Part 2	Prepare the source site (account, web service, permissions, token)
Part 3	Connect the destination course with the setup wizard
Part 4	Optional: share grades, submissions and marks
Part 5	If something goes wrong
Part 6	Final checklist

Written for Course Sync v1.27.0-beta · Moodle 5.0 to 5.3 · Screenshots taken on Moodle 5.1 with the Boost theme. Other versions look very slightly different; the menu names are the same.

READ THIS FIRST

Before you start

Course Sync copies activities (pages, quizzes, assignments and more) from a course on one Moodle site into a course on another. Moodle has no built-in way to hand activities from one site to another, so Course Sync brings its own. That is why **the plugin must be installed on both sites**, not just the one that does the copying.

Source site

The site that *has* the content. An administrator switches on a small web service here, once. Nothing is installed apart from the plugin.

Destination site

The site that *receives* the content. It holds the Course Sync block. Teachers use the block afterwards to pull in changes.

What you need

- **Administrator login on both sites** (you will use *Site administration* a lot).
- **The plugin ZIP file**, named like `block_coursesync-v1.27.0-beta.zip`. Keep it zipped. Do not unpack it.
- **A secure address for the source site.** The address must start with `https://` and have a valid certificate. Course Sync refuses plain `http://` because the secret token would travel unprotected.
- **Moodle 5.0 or newer on both sites.** The two sites may run different versions.

Words used in this guide

Word	Meaning
Web service	A door on the source site that lets another computer ask questions. It is closed until you open it (Part 2).
Sync account	A user account on the source site that Course Sync uses to ask questions. It is not a real person and nobody logs in with it.
Token	A long secret code, like a password for the sync account. You create it on the source site and paste it into the destination site once. Treat it like a password.
Role	A named bundle of permissions in Moodle. You will make two small roles and give them to the sync account.

Tips

Keep the source and destination sites in two labelled browser tabs so you never change the wrong one. Red numbered circles in the screenshots show where to click, in order. The screenshots come from test sites called “CSB” (source) and “CSA” (destination), and some boxes you must tick are already ticked in the pictures. Allow about 30 minutes.

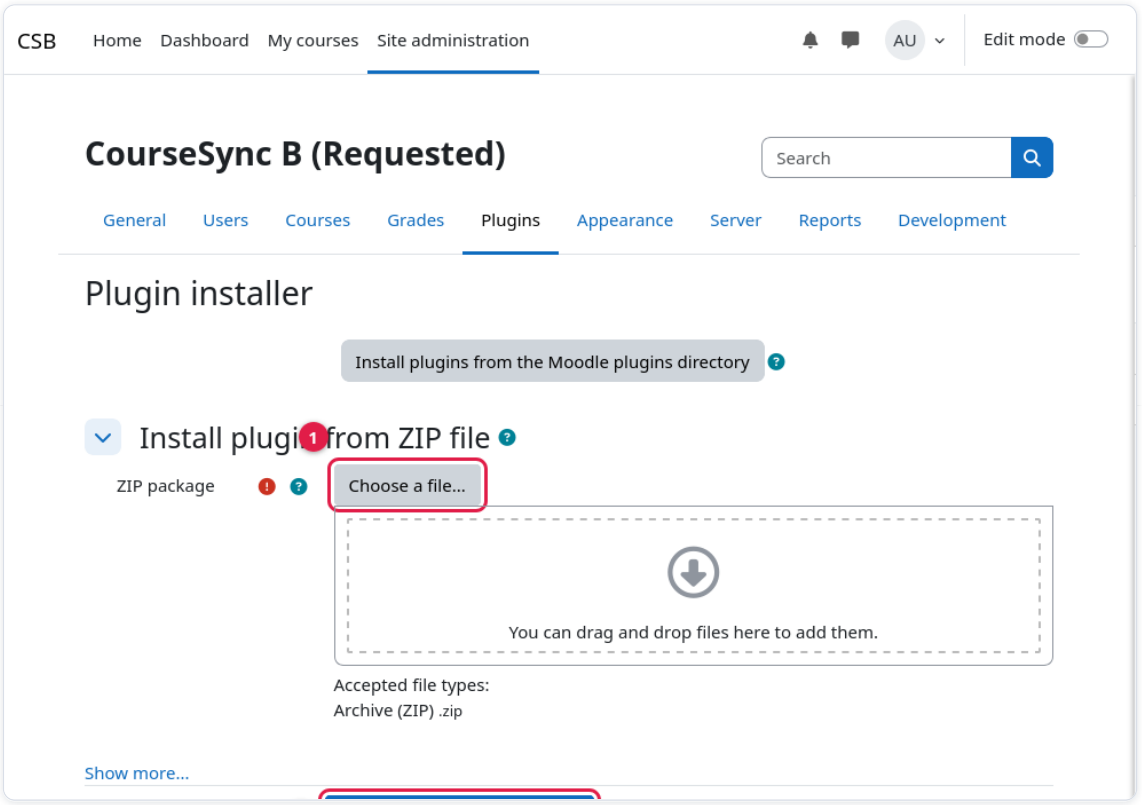
PART 1 · ON BOTH SITES

Install the plugin

1.1 Upload the ZIP file

> **Where: Site administration → Plugins → Install plugins**

1. Sign in as an administrator.
2. Open **Site administration**, choose the **Plugins** tab, then **Install plugins**.
3. Drag the ZIP file onto the grey box (1), or click **Choose a file…** and pick it.
4. Click **Install plugin from the ZIP file** (2).
5. Moodle shows what it is about to install. Click **Continue** and then **Upgrade Moodle database now**. Wait for the green *Success* messages and click **Continue** again.
6. **Repeat on the second site.**



Install plugins. The ZIP goes in the file box (1) and you then click the install button (2).

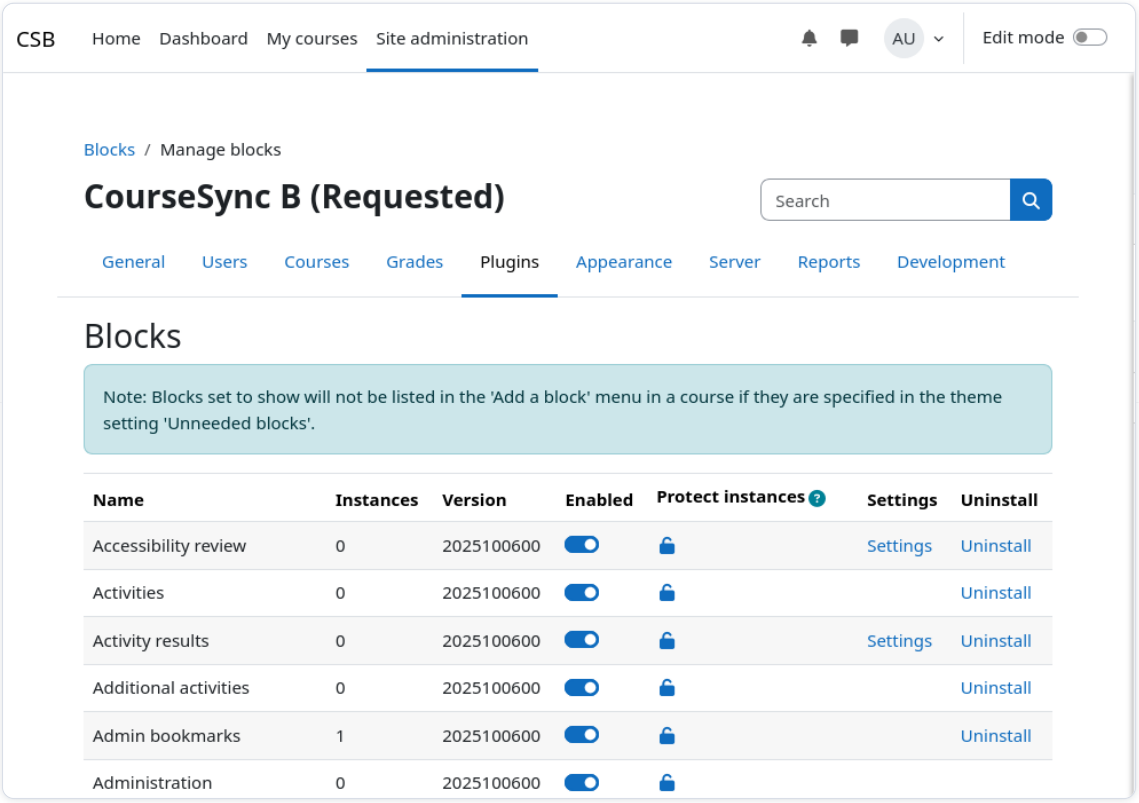
No "Install plugins" page, or the upload is refused?

Some hosts lock this screen so plugins cannot be uploaded from the browser. Ask whoever looks after the server to put the plugin folder in `public/blocks/coursesync` for you. Everything after that is the same as in this guide.

1.2 Check it is there

> **Where: Site administration → Plugins → Blocks → Manage blocks**

Look for **Course Sync** in the list 1. It should be *enabled* (the eye icon is open). Do this on both sites.



Manage blocks lists Course Sync with its version number.

PART 2 · SOURCE SITE ONLY

Prepare the source site

Make sure you are on the source site

Every step in this part happens on the site that *has* the course you want to copy from. Check the site name at the top left of each screenshot and match it with your own source site's name.

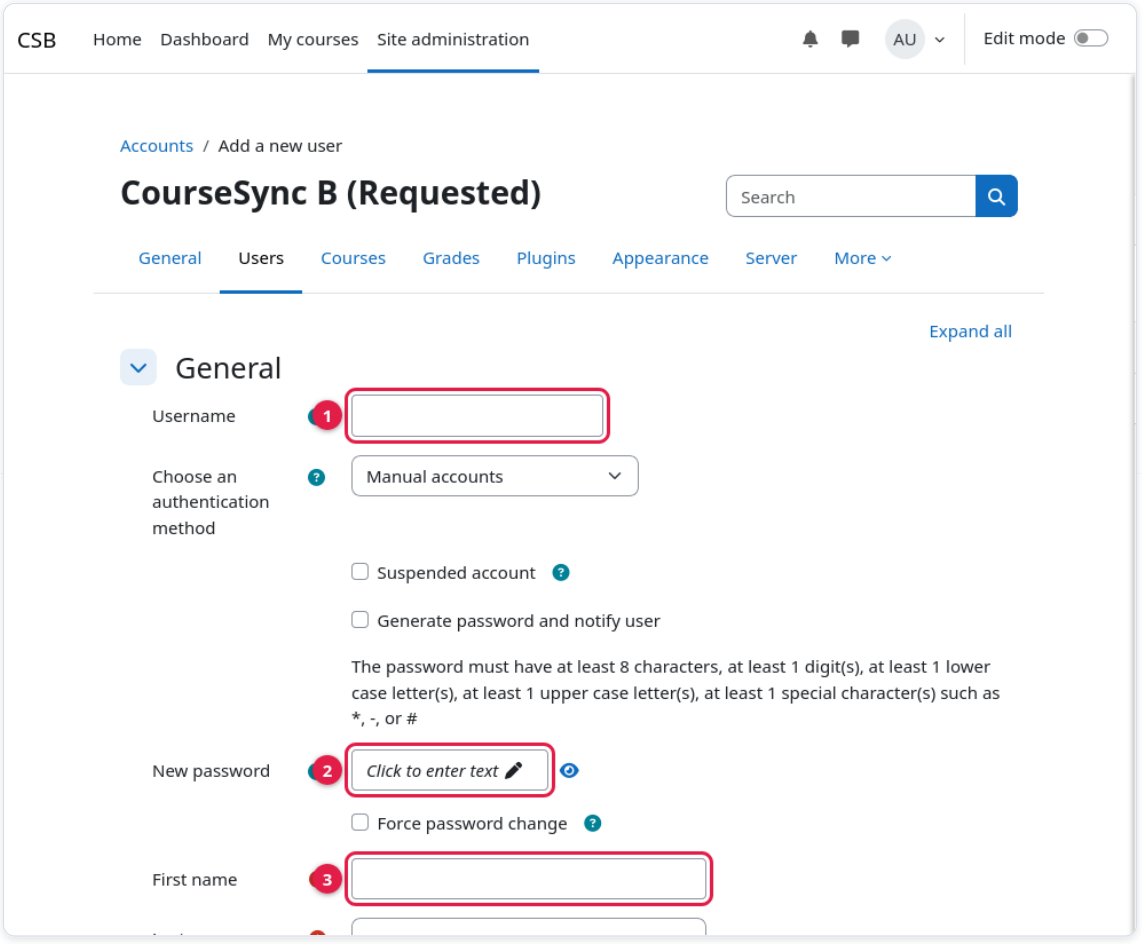
Course Sync never changes another site's settings for you. That is deliberate: letting one site control another is a much bigger decision than a copy needs. So you open the door by hand, in eight short steps.

2.1 Create the sync account

> **Where: Site administration → Users → Accounts → Add a new user**

Create a dedicated account that exists only for Course Sync. A separate account is easy to spot in logs and easy to switch off later.

1. **Username** 1: for example `coursesync_service` (lower-case, no spaces).
2. **New password** 2: choose a strong one. Nobody will type it, so you do not need to remember it. Store it in your password manager.
3. **First name / Surname** 3: for example *Course* and *Sync*.
4. **Email address** 4: any address that works (or an address ending in `.invalid` if you do not want mail going to it).
5. Click **Create user** at the bottom.

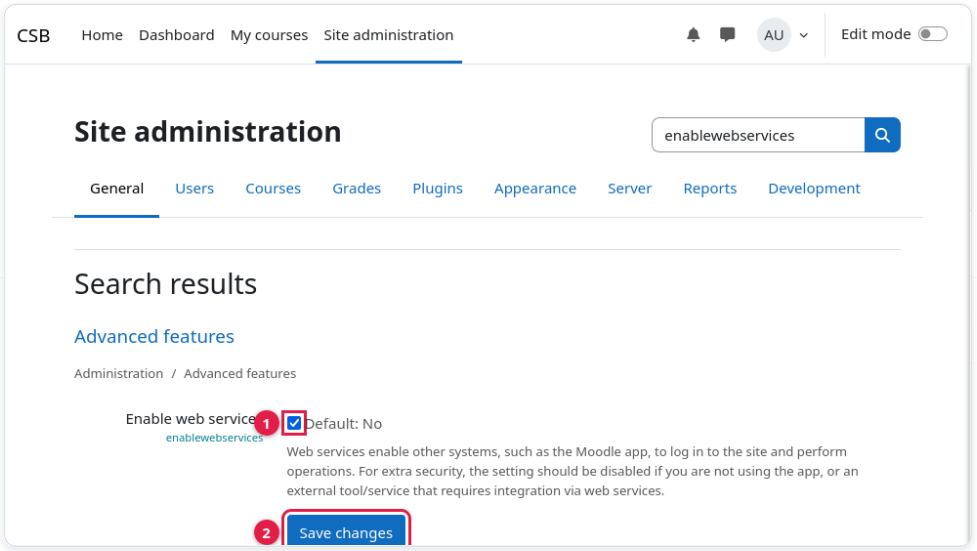


Add a new user. Only the four highlighted boxes matter for the sync account.

2.2 Switch on web services

> **Where: Site administration → General → Advanced features** (you can also type *enablewebservices* in the admin search box)

1. Tick **Enable web services** 1.
2. Click **Save changes** 2.

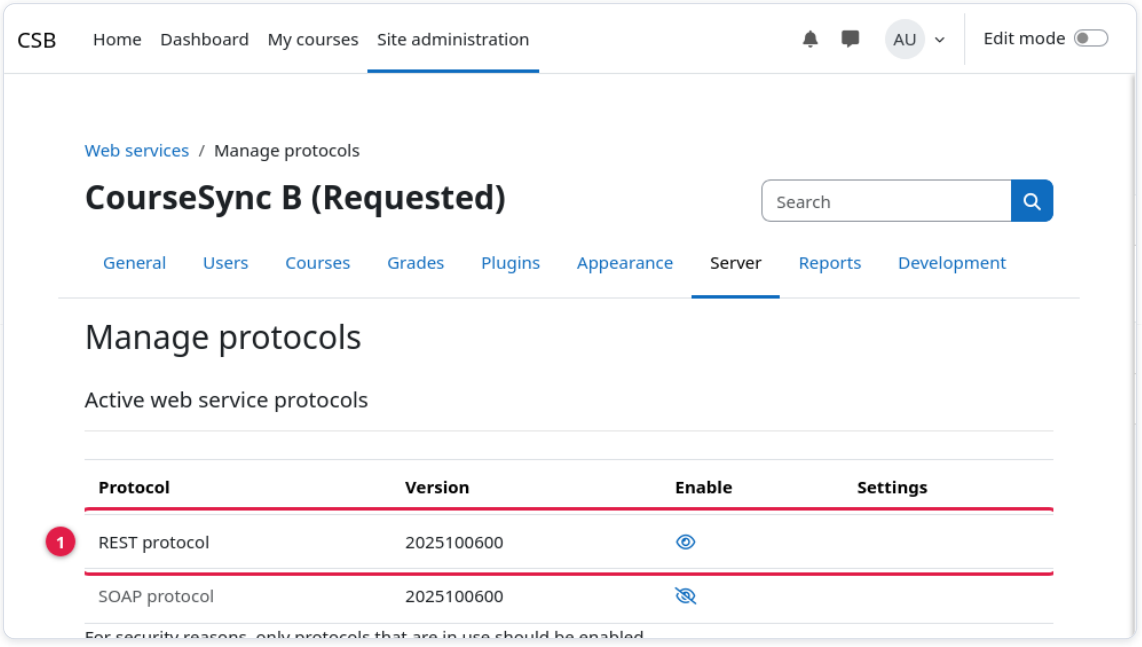


Enable web services, then save.

2.3 Switch on the REST protocol

> **Where: Site administration → Server → Web services → Manage protocols**

Click the **eye icon** in the *Enable* column of the **REST protocol** row 1 until the eye is open (not crossed out). Course Sync only uses REST. Leave SOAP switched off.



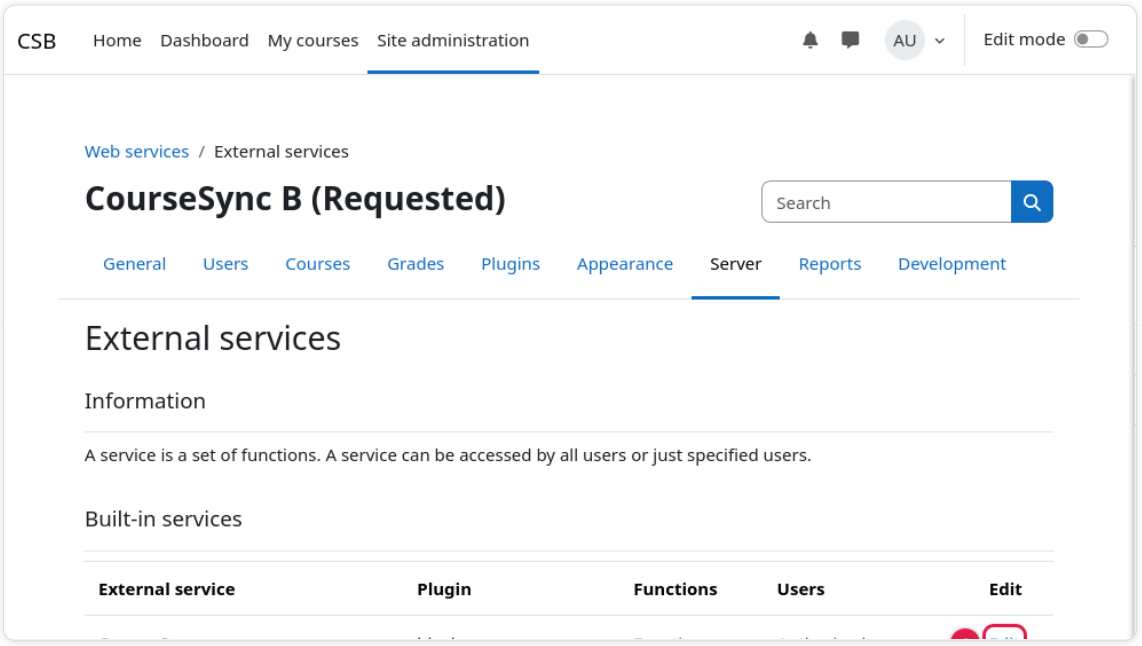
REST protocol enabled. The SOAP row stays crossed out.

2.4 Switch on the Course Sync service

> **Where: Site administration → Server → Web services → External services**

A service called **Course Sync** is already in the list, because the plugin created it. You do not need to build anything.

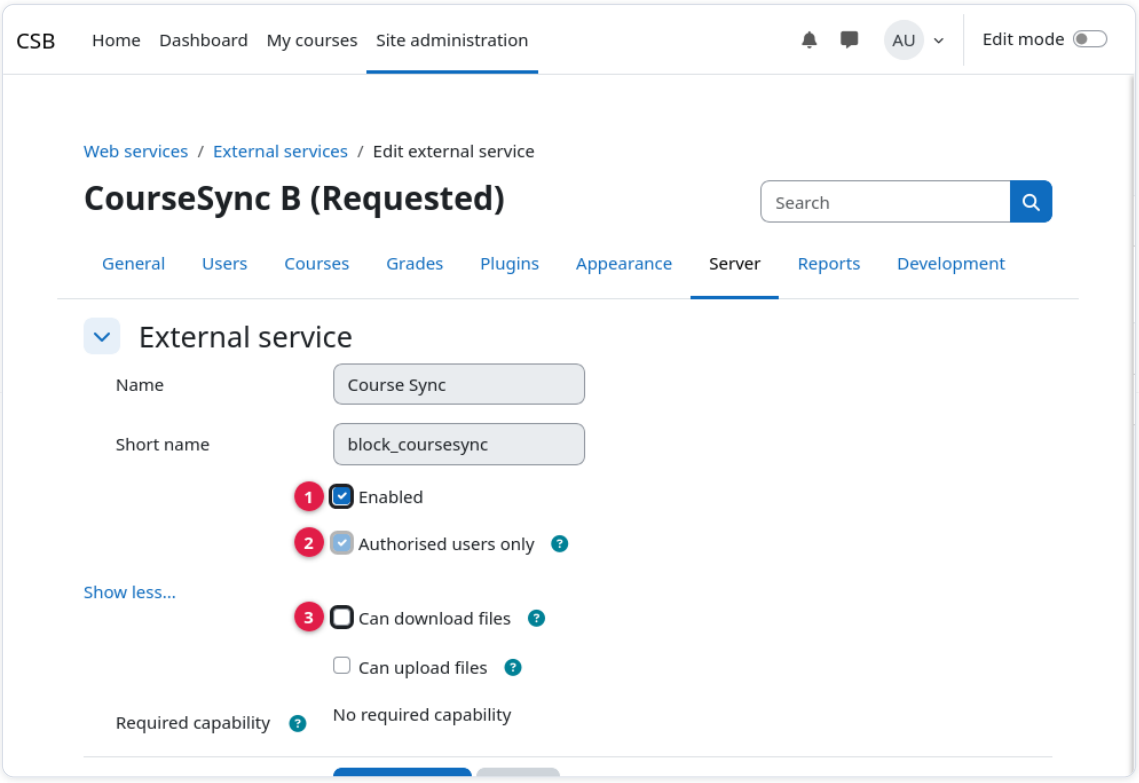
1. On the **Course Sync** row, click **Edit** 1.



External services. “Edit” on the Course Sync row opens its settings.

2.4 (continued) Set the three tick-boxes

1. Tick **Enabled** 1.
2. Leave **Authorised users only** 2 ticked. This means only the accounts you name can use the service.
3. Click **Show more…**. Make sure **Can download files** 3 is **not** ticked.
4. Click **Save changes**.



The finished service settings: Enabled ticked, Authorised users only ticked, Can download files empty.

Leave “Can download files” off

Course Sync has its own, safer way of moving files. Switching this box on would let the token fetch any file the account can reach.

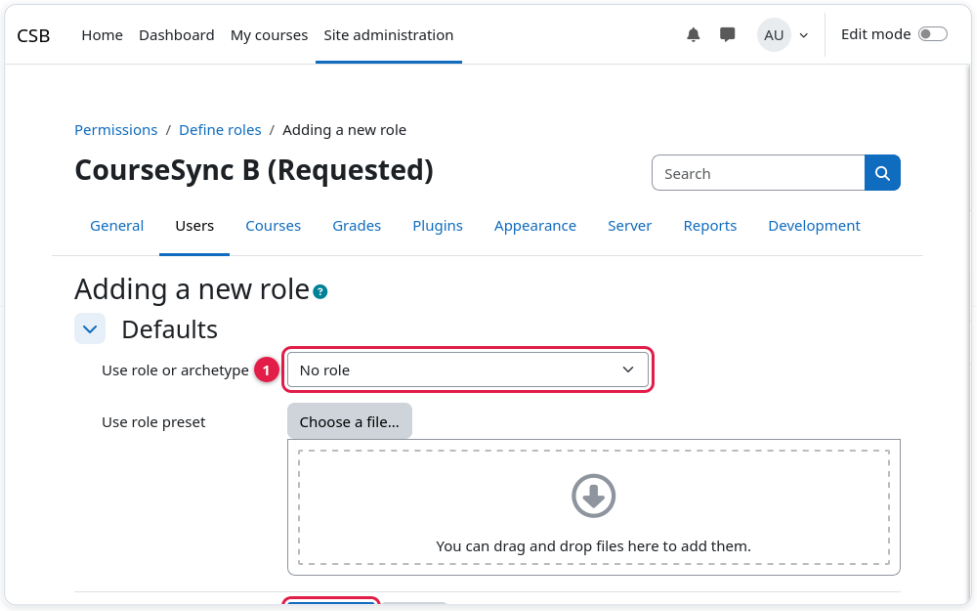
2.5 Create two small roles

> **Where: Site administration → Users → Permissions → Define roles → Add a new role**

The sync account needs three permissions. It is tidiest (and safest) to give them as two small roles. Why two? One permission must apply to the whole site, but the other two should apply *only* to the courses you are willing to share. Whatever the account is allowed to see is exactly what the token can read.

Role to make	Short name	Can be assigned in	Set to Allow
Role 1 "Course Sync REST"	`coursesyncrest`	System	`webservice/rest:use` "Use REST protocol"
Role 2 "Course Sync reader"	`coursesyncreader`	Category and Course	`block/coursesync:sync` "Trigger a course sync" `moodle/course:view` "View courses without participation"

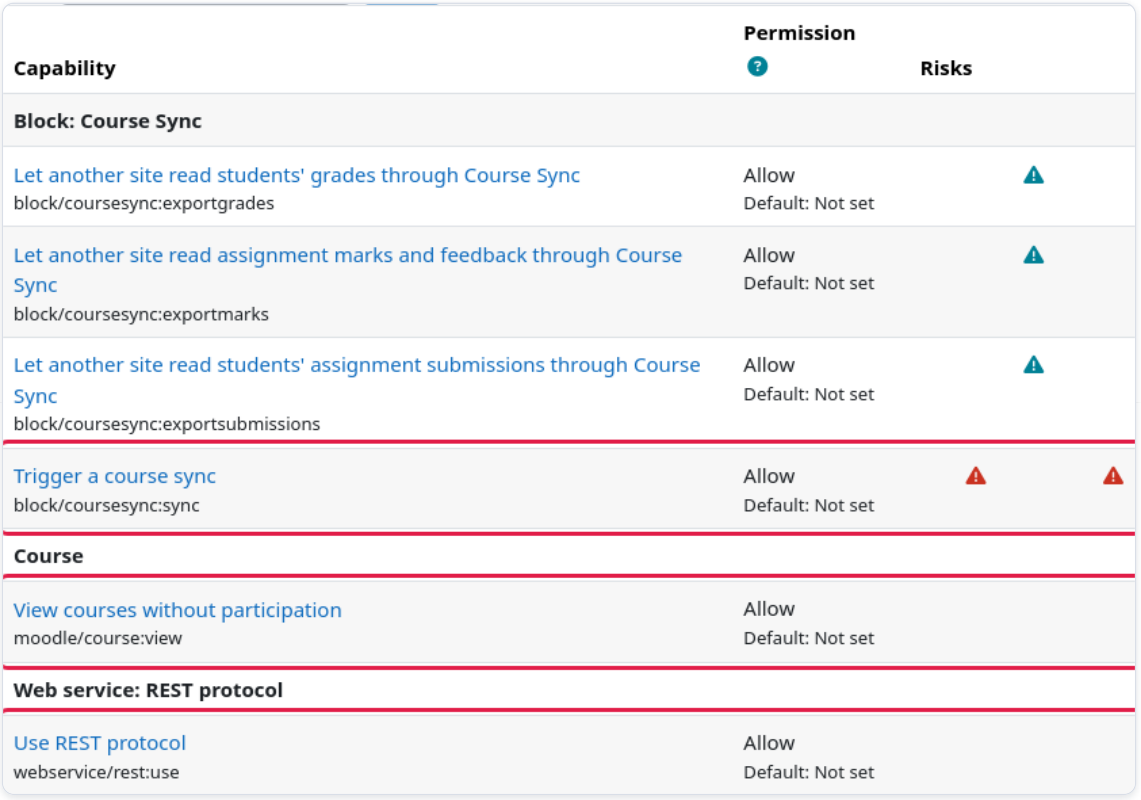
1. On *Add a new role*, leave **Use role or archetype** on *No role* (1) and click **Continue** (2).
2. Fill in the **Short name** and **Custom full name** from the table.
3. Under **Context types where this role may be assigned**, tick the boxes from the table (*System* for Role 1; *Category* and *Course* for Role 2).
4. In the **Filter** box above the capability list, type part of the capability name (for example *rest:use*) and tick **Allow** in its row.
5. Scroll to the bottom and click **Create this role**.
6. Repeat for Role 2.



Adding a new role starts here.

2.5 (continued) What a finished role looks like

After you save, open the role to double-check it. The picture shows a finished role. The three red-outlined rows are the permissions Course Sync needs. Yours will list only the ones you ticked.



Capability	Permission	Risks
Block: Course Sync		
Let another site read students' grades through Course Sync block/coursesync:exportgrades	Allow Default: Not set	
Let another site read assignment marks and feedback through Course Sync block/coursesync:exportmarks	Allow Default: Not set	
Let another site read students' assignment submissions through Course Sync block/coursesync:exportsubmissions	Allow Default: Not set	
Trigger a course sync block/coursesync:sync	Allow Default: Not set	
Course		
View courses without participation moodle/course:view	Allow Default: Not set	
Web service: REST protocol		
Use REST protocol webservice/rest:use	Allow Default: Not set	

A finished role (“Viewing the definition of role”). The test role in this picture also holds the optional sharing permissions from Part 4, so it lists more rows than yours will.

Why these three permissions?

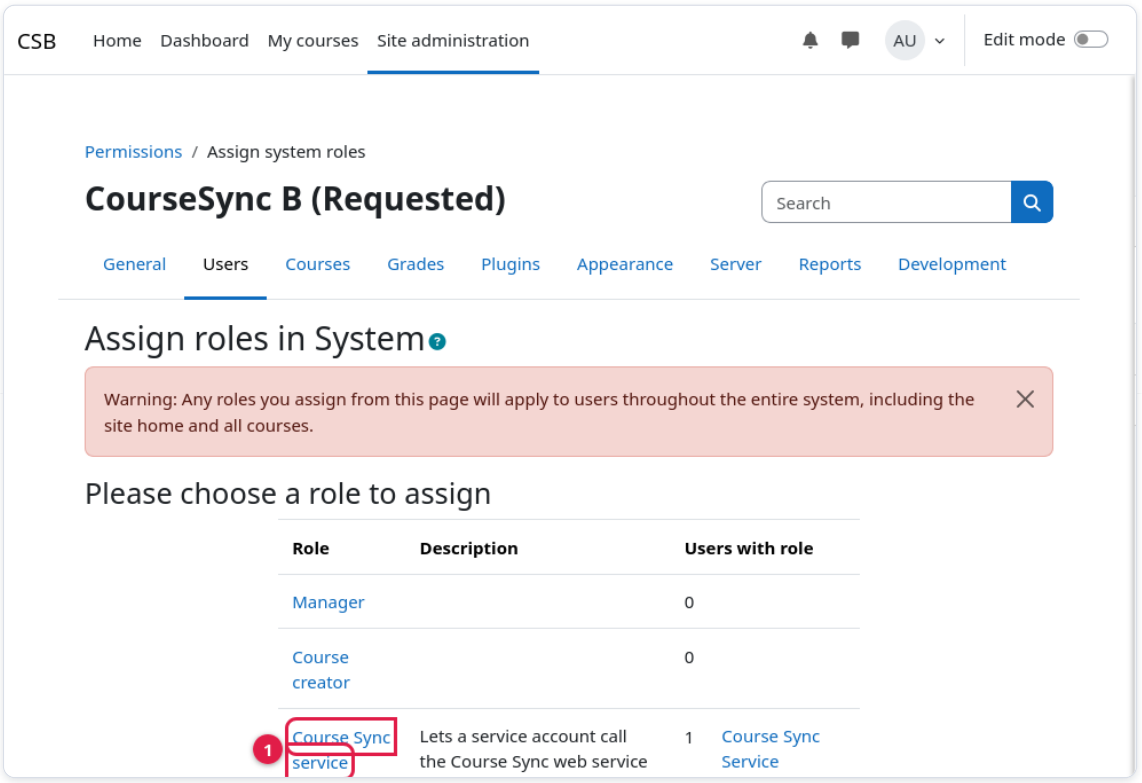
`webservice/rest:use` is Moodle’s permission to call anything over REST. No role has it by default, so forgetting it is common. `block/coursesync:sync` is Course Sync’s own permission. `moodle/course:view` lets the account read a course it is not enrolled in; the account is never a course participant, so without it every request fails. **Missing the third one is the most common set-up mistake.**

2.6 Give the account its roles

(a) Role 1 across the whole site.

> **Where: Site administration → Users → Permissions → Assign system roles**

1. Click your new **Course Sync REST** role (1).
2. Pick the sync account in the right-hand list and click **Add**.

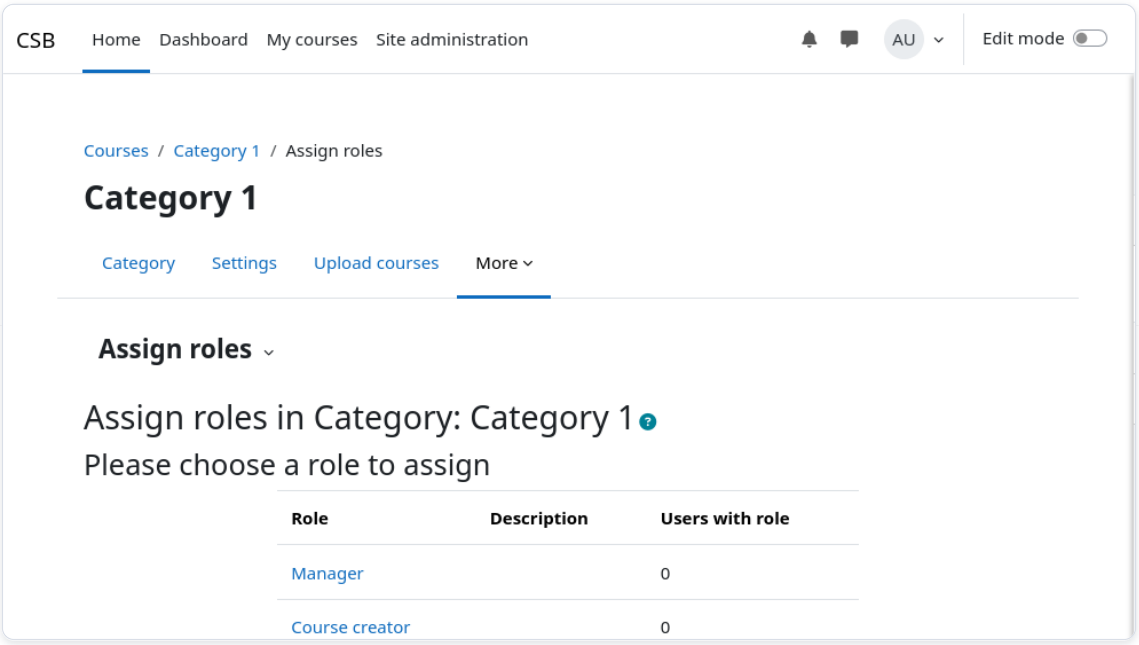


Assign system roles. Pick the role, then add the sync account.

(b) Role 2 only where it may read.

> **Where:** **Site administration → Courses → Manage courses and categories**, click the category, then **More → Assign roles** (or open the single course and use **Participants → Enrolment methods**'s sibling, **More → Assign roles**)

1. Choose the **category** that holds the course(s) you want to copy *from*. Using one course is even tighter.
2. Click your **Course Sync reader** role and add the sync account.



Assign roles in a category. The sync account can read only the courses inside this category.

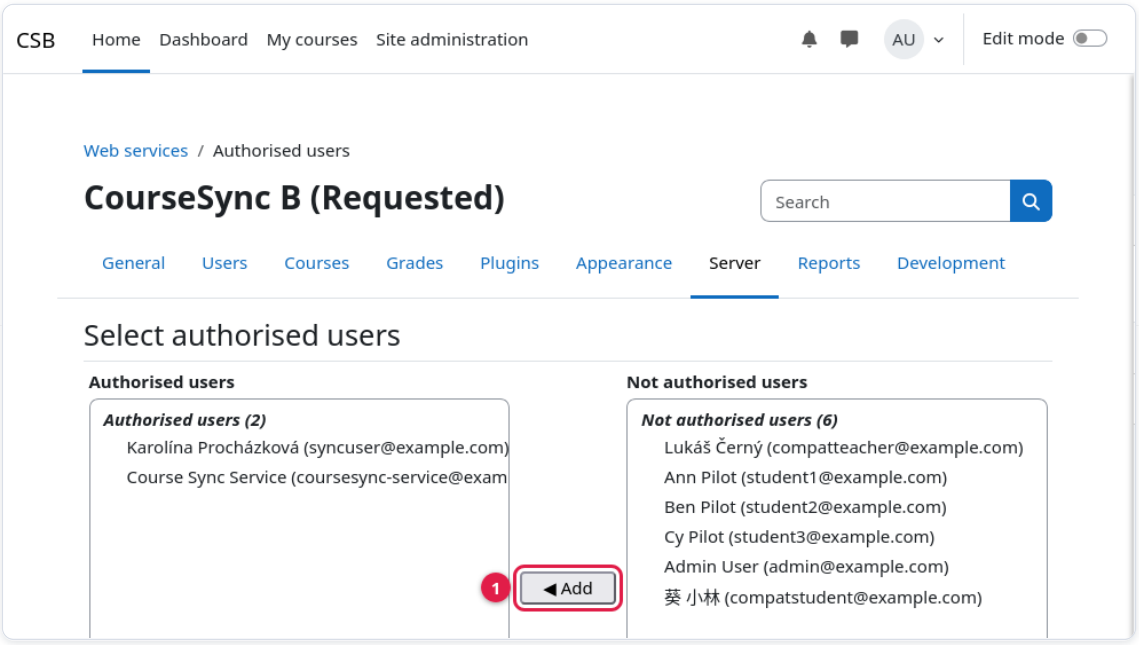
Where you give this role is what the token can read

With this role the account sees *everything* in the course, including hidden activities and quiz answers. If you gave it at site level it could read every course on the site. Keep it as narrow as you can. Use a separate account and token for each destination that should see something different.

2.7 Allow the account to use the service

> **Where: Site administration → Server → Web services → External services** → on the Course Sync row click **Authorised users**

1. Select the sync account in the right-hand list (*Not authorised users*).
2. Click ◀ **Add** (1). The name moves to the left (*Authorised users*).



Authorised users. Accounts on the left may use the Course Sync service.

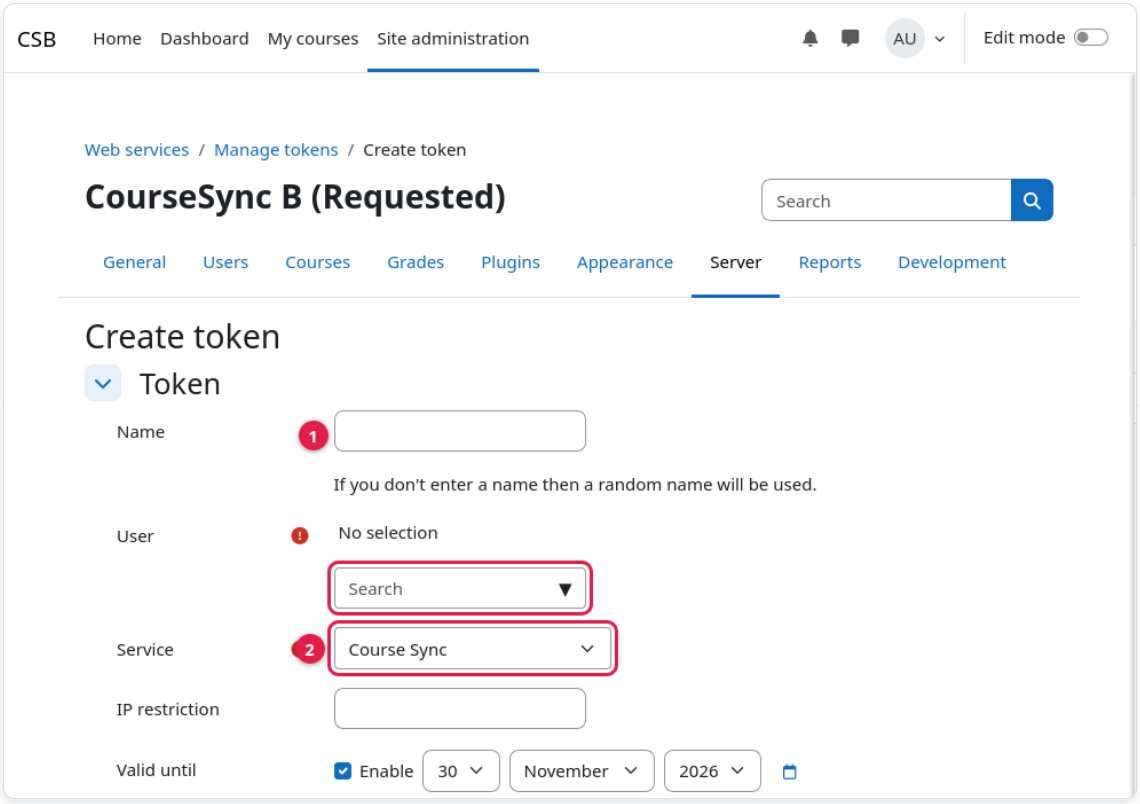
A warning here is normal

Moodle may show a warning that the account lacks `block/coursesync:sync` or `moodle/course:view`. It only checks site level, and you gave those two permissions in a category on purpose, so you can ignore it. A warning about `webservice/rest:use` is *not* normal: go back to 2.5 and 2.6(a).

2.8 Create the token

Where: Site administration → Server → Web services → Manage tokens → Create token

1. **User** 1: start typing the sync account's name and pick it.
2. **Service** 2: choose **Course Sync**.
3. **Valid until**: leave it enabled with a date that suits you, or untick *Enable* for no expiry. The connection stops working on that date.
4. Click **Save changes** 3.
5. Moodle shows the new token in the list. **Select the long code and copy it** (Ctrl+C on Windows and Linux, Cmd+C on a Mac). You will paste it in Part 3.



Create token. User, service, save.

Treat the token like a password

Anyone holding it can read what the sync account can read. Do not email it or paste it into a chat. Paste it straight into the wizard in Part 3.

The source site is ready

You now have: a sync account, web services on, REST on, the Course Sync service enabled and authorised, two roles assigned, and a token on your clipboard. Switch to the **destination** site.

PART 3 · DESTINATION SITE

Connect a course

Who can do this part

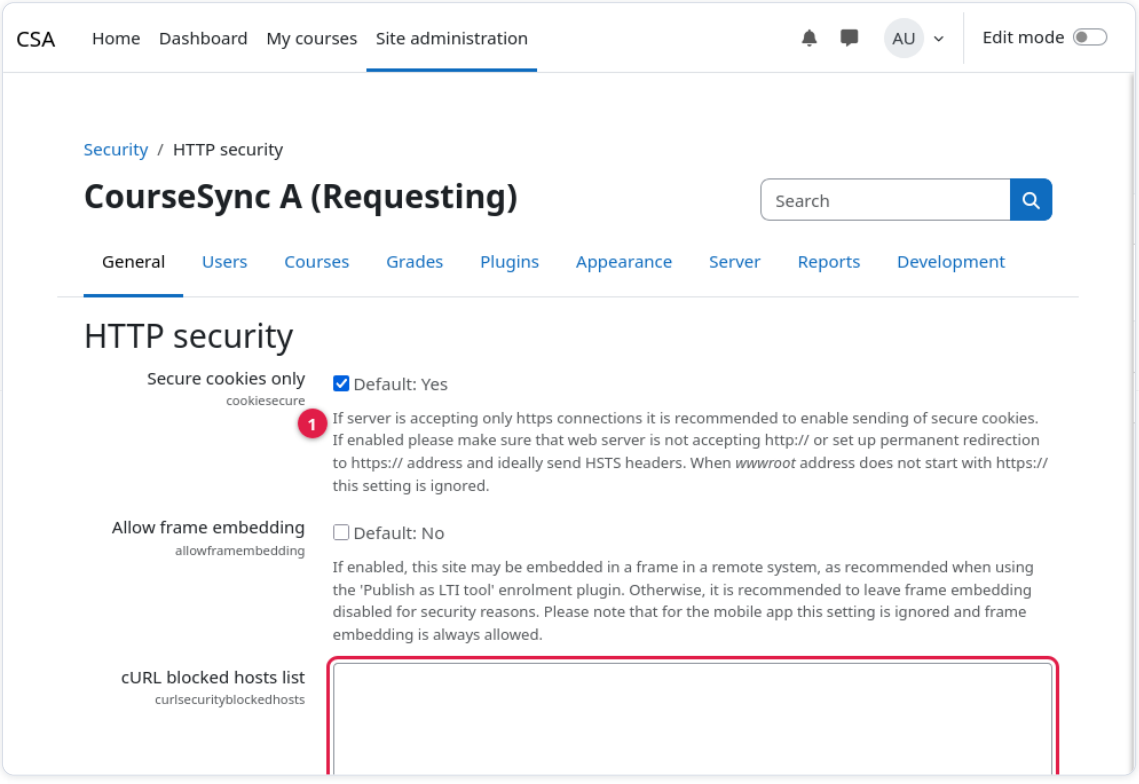
The setup wizard needs the permission `block/coursesync:configure`, which only administrators and managers have by default. That is intentional: the connection decides what this site may read on the other one. Once a course is connected, its teachers can sync it themselves.

3.1 Let this site reach the source site

Where: Site administration → General → Security → HTTP security

Moodle filters the web addresses its own server may contact. If your source site uses a normal web address (port 443) you can skip this step. Check these two settings if the source is on a different port, or if you later see *"This site is not allowed to make connections to that address"*:

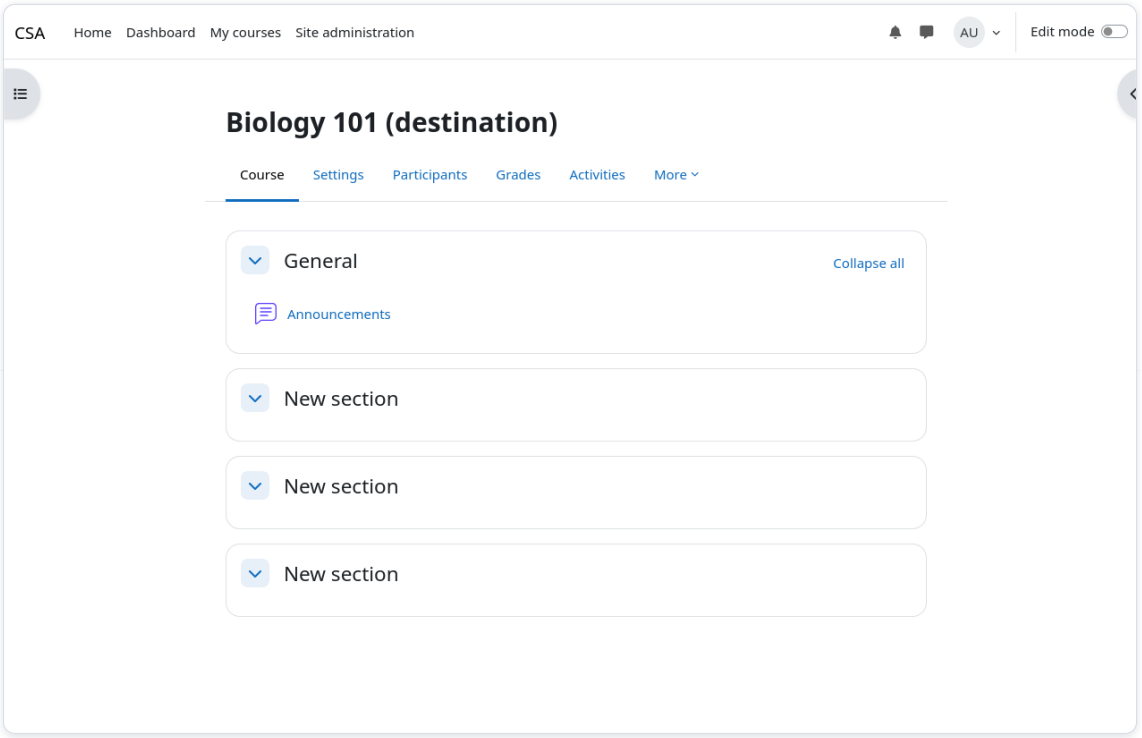
1. **cURL blocked hosts** 1: must **not** list the source site's address.
2. **cURL allowed ports** 2: must include the source site's port. The default allows only 80 and 443. Add the other port on a new line, then click **Save changes**.



HTTP security. The two highlighted settings decide whether this site may contact the source.

3.2 Put the Course Sync block on the course

1. Open the **destination course** (the one that should receive activities).
2. Switch **Edit mode** on (top right).
3. Open the **block drawer** (arrow on the right edge) and click **Add a block**. Choose **Course Sync**.
4. Switch Edit mode off again. The block shows *"Course Sync — not yet configured"* with a **Set up the connection** button 1.

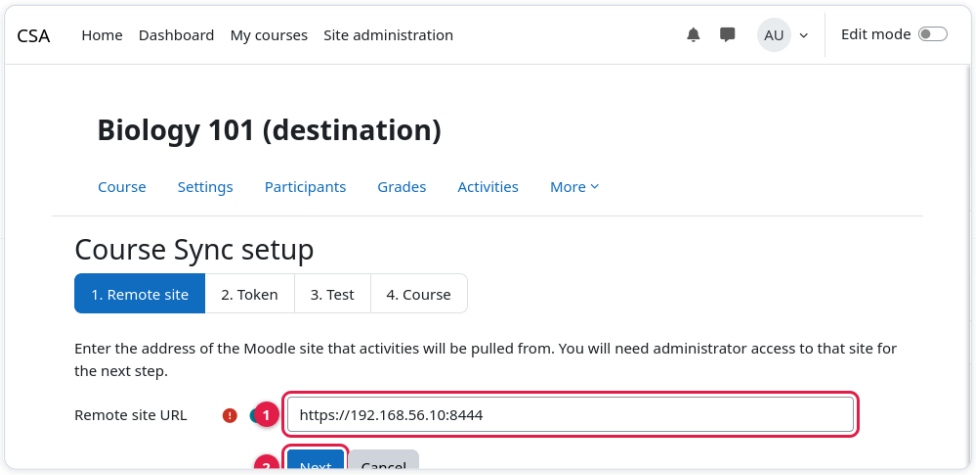


A new Course Sync block, waiting to be set up. Click the button.

3.3 Run the setup wizard

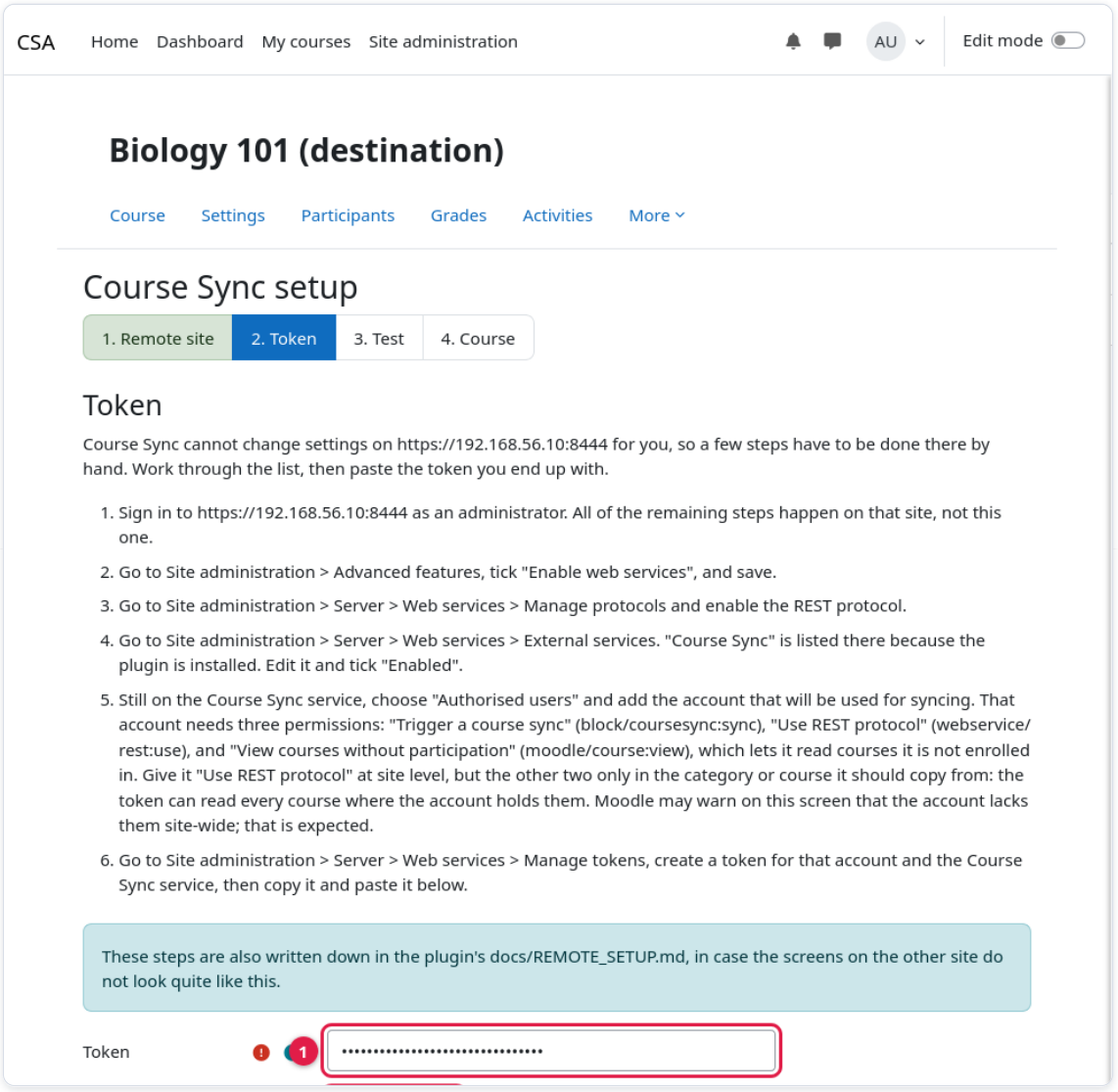
The wizard has four tabs along the top (*1. Remote site, 2. Token, 3. Test, 4. Course*). The finished steps turn green.

Step 1 · Remote site. Type the source site's address, exactly as you open it in your browser but **without** anything after the site name. For example `https://moodle.example.edu`, not `https://moodle.example.edu/my/`. Click **Next** 2.



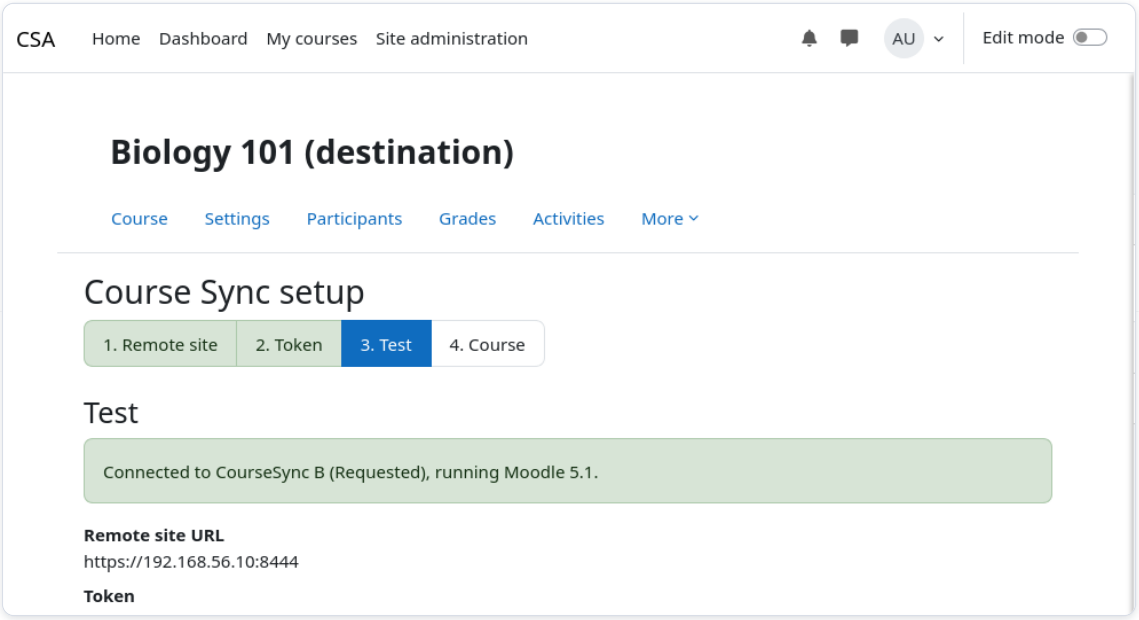
Step 1: the address of the source site.

Step 2 · Token. The page repeats the source-site steps you already did in Part 2 as a checklist. Paste the token you copied into the **Token** box 1 (it shows as dots) and click **Save and test** 2.



Step 2: paste the token and click “Save and test”. The token is encrypted before Moodle stores it.

Step 3 · Test. Course Sync now contacts the source site. A working connection reports the source site's name and Moodle version, for example *"Connected to CourseSync B (Requested), running Moodle 5.1"*. Click **Next** 2. You can click **Test the connection** 1 again at any time.

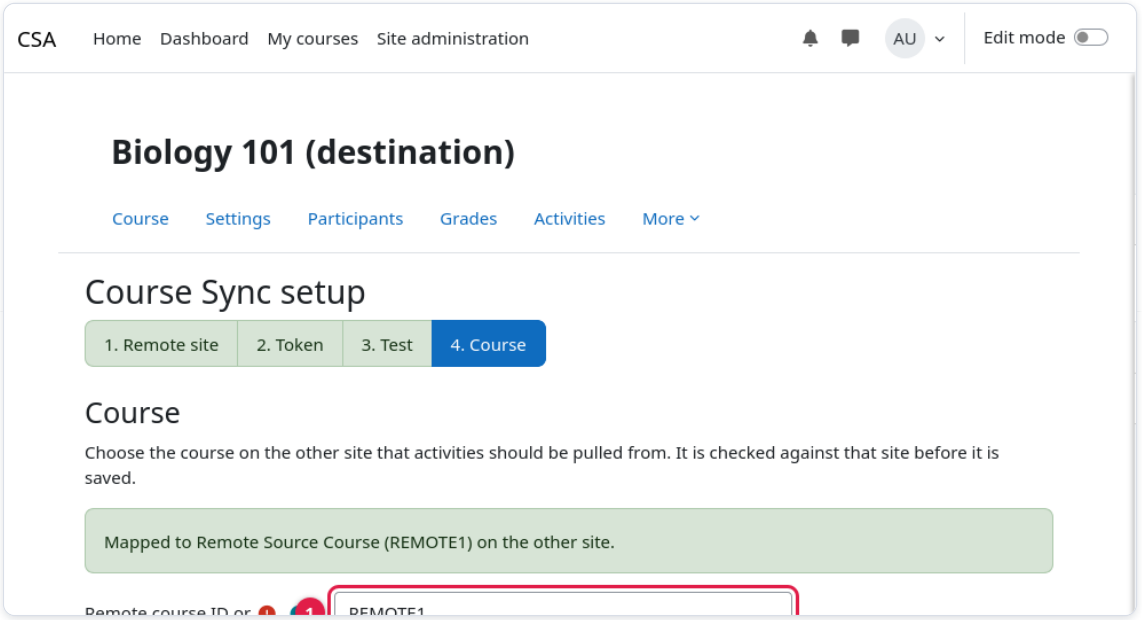


Step 3: a successful test shows the other site's name.

The test failed?

Do not guess. The message names the cause. Go to Part 5 for the fix for each message.

Step 4 · Course. Type the **short name** or the **ID number** of the course on the *source* site that activities should come from 1, then click **Save the course** 2. Course Sync checks it with the source site before saving. A green box says *"Mapped to …"* with the course's full name.



Step 4: which course on the source site to copy from.

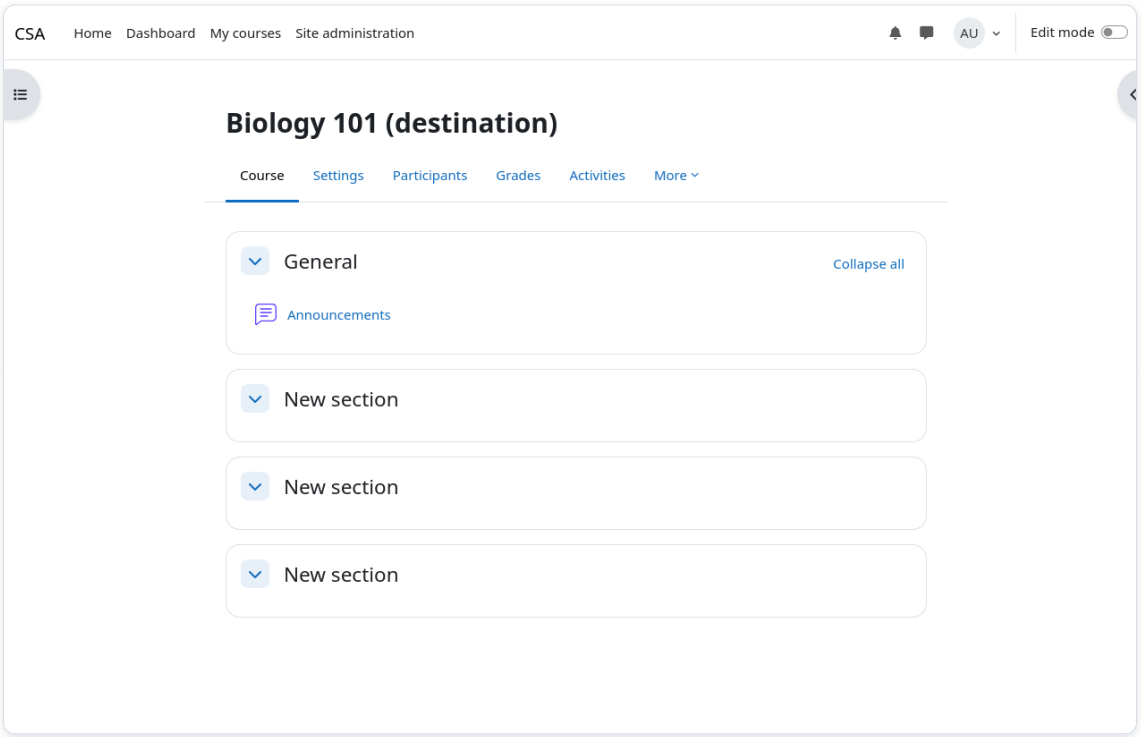
Where do I find the short name?

On the source site, open the course and look at **Settings**: **Course short name** is the second box. The ID number in the address bar (`…/course/view.php?id=12`) also works.

3.4 See it working

Back on the course page the block now shows a green **Connected to …** line, the course it pulls from, and its buttons:

Button	What it does
Check Now	Looks for new or changed activities on the source course.
See Changes	Shows what would be copied, *before* anything is copied.
Pull Grades / Submissions / Marks and Feedback	Optional. Only work if you switch them on in Part 4.
View sync history	A log of every sync: when, who, and what was copied.
Manage the Connections	Re-open the wizard to change the address, token or course.



A connected block. Click 1 Check Now to run the first sync.

Done

Click **Check Now**, then **See Changes** to preview what will arrive, and follow the on-screen prompts. Teachers can read the separate *Teacher Guide* for the day-to-day use of the block.

PART 4 · OPTIONAL

Sharing grades, submissions and marks

Copying activities never moves anyone's personal data between sites. Three optional features do, and each is **off** until you switch it on, on **both** sites:

Feature	What moves across
Grades	Each student's gradebook grade and feedback; for quizzes, also their finished attempts and per-question marks.
Assignment submissions	The latest handed-in text and files for each student.
Assignment marks and feedback	The mark and written comment for the latest handed-in attempt.

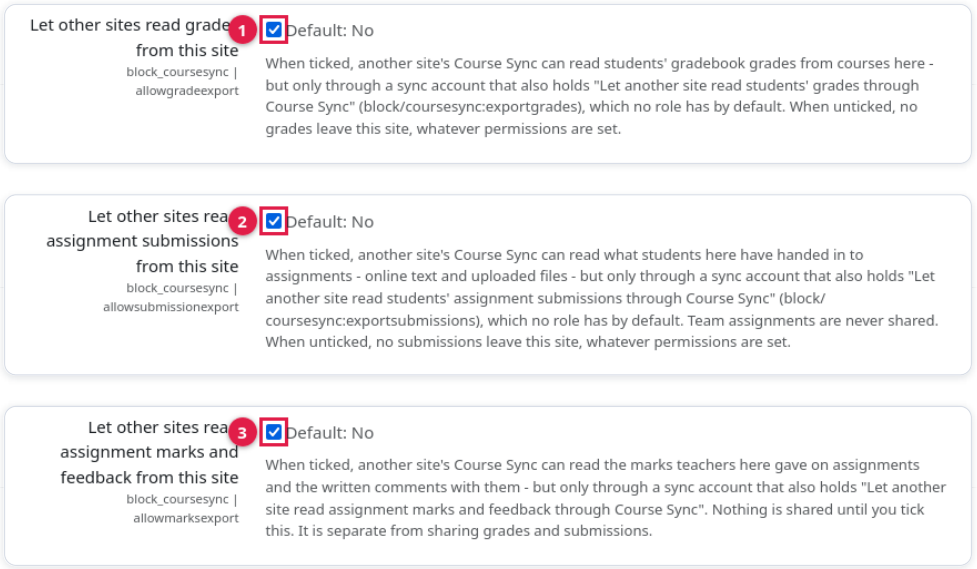
Check usernames first

Students are matched by **username only**. Nobody is created or enrolled, and a username that means two different people on your two sites would give one person the other's data. Only switch these on if the same username is the same person on both sites (usually because both sites take accounts from the same directory or student system).

4.1 On the source site: allow sharing

> **Where: Site administration → Plugins → Blocks → Course Sync**

1. Tick the “**Let other sites read …**” box for each feature you want to share (the three boxes outlined in red below).
2. Click **Save changes**.
3. Edit the **Course Sync reader** role (Part 2.5) and set **Allow** for the matching permission: `block/coursesync:exportgrades`, `…:exportsubmissions` or `…:exportmarks`. **No role has these by default**, not even manager.



The three “read from this site” switches on the source site: grades, assignment submissions, assignment marks and feedback. The same page has a matching “pull into this site” switch for each, used on the destination.

4.2 On the destination site: allow pulling

Open the same settings page on the **destination** and tick the matching "**Let teachers pull … into this site**" boxes. Then check that teachers hold the right permissions in the course:

Feature	Teachers need
Grades	`block/coursesync:pullgrades` (editing teachers and managers have it) and `moodle/grade:edit`
Submissions	`block/coursesync:pullsubmissions`, `mod/assign:grade` and `mod/assign:editothersubmission`. **No role has the last one by default**; grant it to the role that should pull, or only administrators will see the button.
Marks and feedback	`block/coursesync:pullmarks` and `mod/assign:grade`

Safe by design
Every pull shows a preview first. Course Sync never replaces a grade, mark or submission that someone already entered on the destination.

PART 5 · TROUBLESHOOTING

If something goes wrong

The wizard's **Test the connection** button tells you what is wrong in plain words. Find the message in the left column.

What the block says	What it usually means, and the fix
The other site could not be reached	The address is wrong, the source site is down, or a firewall sits between the two. Check the address in a browser from a different computer.
Nothing that looks like a Moodle site answered	The address has something extra on the end (like `/login`), or it points to a different website. Use only `https://sitename`.
The security certificate could not be verified	The source site's HTTPS certificate is self-signed, expired or made for another name. It must be a valid certificate.
The security token was not accepted	The token was mistyped, expired or deleted. Create a new one (Part 2.8) and paste it again, with **Change the token**.
The Course Sync service is not switched on	Step 2.4 was missed, or the token belongs to a different service. Make sure the token's *Service* says **Course Sync**.
The account is not allowed to use Course Sync	Step 2.5, 2.6 or 2.7 was missed: check the three permissions and that the account is in *Authorised users*.
No course matches that id or shortname	A typo, or the sync account has no sync permission in that course (Part 2.6b).
The account is not allowed to see that course	`moodle/course:view` is missing from the reader role.
Course Sync is not installed on the other site	Part 1 was only done on one site.
This site is not allowed to make connections to that address	Part 3.1: the HTTP security settings on the destination block the source's address or port.
The saved token could not be read	The site's secret-key file could not be read by the web server (it was created by another account), or the database was restored without it. Create the token again and re-enter it; if it persists, ask your server administrator to check the file permissions on `secret/key` in the Moodle data folder.
The other site does not share grades / the account is not allowed to read grades	**Pull grades** only. Part 4: the switch on the source is off, or the permission is missing in *the course you copy from*.
The other site's Course Sync is too old to share grades	Upgrade the plugin on the source site (Part 1).

Changing things later

You can re-open the wizard any time from **Manage the Connections** in the block. To stop a site reading from the source, delete its token on the source at **Site administration → Server → Web services → Manage tokens**. Uninstalling the plugin does *not* delete tokens, so do that yourself.

PART 6 · CHECKLIST

Final checklist

Both sites

- [ ] Plugin ZIP installed; **Course Sync** appears in Manage blocks.
- [ ] Both sites reachable over `https://` with valid certificates.

Source site

- [ ] Sync account created.
- [ ] **Enable web services** ticked.
- [ ] **REST protocol** enabled.
- [ ] Course Sync service: **Enabled** on, **Authorised users only** on, **Can download files** off.
- [ ] Role 1 (`webservice/rest:use`) assigned at site level.
- [ ] Role 2 (`block/coursesync:sync` + `moodle/course:view`) assigned in the category/course to copy from.
- [ ] Sync account listed under **Authorised users**.
- [ ] Token created for the sync account and the **Course Sync** service.

Destination site

- [ ] HTTP security allows the source's port and does not block its host.
- [ ] Course Sync block added to the course.
- [ ] Wizard steps 1 to 4 done; step 3 reported the source site's name.
- [ ] Block shows **Connected to …** and the right source course.
- [ ] First **Check Now** and **See Changes** run.

Only if you share grades, submissions or marks

- [ ] Usernames mean the same people on both sites.
- [ ] "Let other sites read …" ticked on the source; matching export permission on the reader role.
- [ ] "Let teachers pull …" ticked on the destination; teachers hold the permissions in Part 4.2.

Need more detail?

The plugin's `docs` folder holds the longer guides: `INSTALL.md` (administrators), `REMOTE_SETUP.md` (source site), `TEACHER_GUIDE.md` (daily use) and `SECURITY.md` (how the connection is protected).